EncoderX-14 Reference Manual

14-bit contactless magnetic absolute angle encoder with I²C interface

Document ENCX14-RM, revision 1.2

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The most frequently referenced items are:

- Register CONF2 and the FILT[2:0] noise filter ..... page 31
- Output noise versus FILT setting (table 31-2) ..... page 31
- Read latency and phase lag (table 12-1) ..... page 12
- Errata E-3: CONF2 cleared by every reset ..... page 58

Revision history and conventions

Revision history

Rev	Date	Changes
1.0	2026-03-02	First release.
1.1	2026-06-18	Added table 12-1 (phase lag at the host control rate). Corrected tAIR maximum in table 20-1.
1.2	2026-10-05	Added errata E-3, E-4 and E-5 (pages 58-59). Clarified that STATUS.RST is sticky (page 33). Added the "rewrite CONF2 after reset" guidance (page 45).

Conventions used in this manual

Numbers. Hexadecimal values carry a 0x prefix (0x4C). Binary values carry a 0b prefix (0b011). All other numbers are decimal.

Bit fields. A field is written NAME[msb:lsb]. FILT[2:0] is the three-bit field FILT that occupies bits 2, 1 and 0 of its register. Bit 0 is the least significant bit.

Register access types.

R	read only
W	write only (reads return 0)
RW	read and write
RC	read, cleared by the read (sticky flag)
OTP	one-time programmable; the value survives power cycles

Angles. One LSB of the angle output is 360°/16384 = 0.02197° = 383.5 µrad (2 x pi / 16384 rad). "Code" means the unsigned 14-bit output value in the range 0 to 16383.

Noise. Noise figures are root-mean-square (RMS) values measured over 10,000 consecutive reads with the magnet at rest, unless stated otherwise.

Typical and limit values. "Typ" values are at 25 °C and VDD = 3.3 V. "Min" and "Max" values hold over the full recommended operating range (page 10).

Notes. Paragraphs headed "Important" describe behaviour that has caused field problems. Read them before writing firmware.

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4 Safety and usage notes

The EncoderX-14 is a fictional component described for software and documentation research. The notes below are written as a real manual would write them, so that the document reads realistically.

4.1 Intended use

The EncoderX-14 measures the absolute angle of a diametrically magnetised magnet that rotates above the package. Typical uses are motor commutation, robot joint position feedback, knobs and gimbals.

The device is not rated for safety functions. A safety-rated joint must use an independent second channel and a diagnostic coverage analysis that does not rely on this device alone.

4.2 Behaviour after a reset

Important: the device has several reset sources (power-on, brown-out, watchdog and soft reset, see page 11 and page 51). After any reset the volatile configuration returns to its default values. A host that writes its configuration only once at start-up will silently lose that configuration after the first brown-out. See errata E-3 on page 58 and the recommended initialisation on page 45.

4.3 Strong magnetic fields

Fields above 120 mT do not damage the device but drive the Hall front end into saturation. The MAGH flag (page 33) is set and the angle output is not valid while MAGH is set.

4.4 Electrostatic discharge

The device is ESD sensitive. Handle it at an ESD-protected workstation. ESD ratings are on page 56.

4.5 Motor drives

Place the encoder at least 15 mm away from motor windings and high-current traces, or shield it. Stray fields from a motor add a slowly varying angle error that the FILT filter cannot remove, because the filter only acts on broadband noise above its corner frequency (page 31).

5 Features

- 14-bit absolute angle output, 16384 codes per revolution
- 1 LSB = 0.02197° (383.5 µrad)
- Contactless: on-axis measurement of a diametric magnet, 30 to 90 mT
- Internal sampling at 8 kHz with a CORDIC angle engine
- Programmable first-order digital noise filter, FILT[2:0], with eight settings from "off" to a 16 ms time constant (page 31)
- Output noise 12 LSB RMS with the filter off, down to 0.75 LSB RMS with the strongest setting (table 31-2)
- I²C interface, standard mode, fast mode and fast-mode plus (up to 1 MHz)
- Two selectable I²C addresses, 0x4C or 0x4D (ADDR pin)
- Programmable zero position and angular range (ZPOS, MPOS)
- Selectable direction of rotation (DIR bit)
- Output hysteresis of 0, 1, 2 or 3 LSB (HYST field)
- Magnet diagnostics: field too weak (MAGL), field too strong (MAGH), magnet detected (MD), automatic gain control readout (AGC)
- Sticky reset flag (STATUS.RST) that reports any reset since the last STATUS read
- Internal watchdog that resets the device if the oscillator stalls
- Supply 3.0 V to 3.6 V, typical current 6.5 mA
- Power-on reset (VPOR 2.8 V) and brown-out reset (VBOR 2.6 V)
- Operating temperature -40 °C to +125 °C
- SOIC-8 package, 4.9 mm x 3.9 mm
- One-time programmable (OTP) memory for ZPOS, MPOS and CONF1

5.1 What the device does not do

- It does not store CONF2 in OTP. The FILT, DIR, HYST and WD fields are volatile and return to their defaults after every reset (page 44).
- It does not provide a velocity output. Velocity must be estimated by the host from successive angle reads (page 50).
- It does not provide an incremental (ABI) or PWM output. All data is read over I²C.

6 Functional block diagram

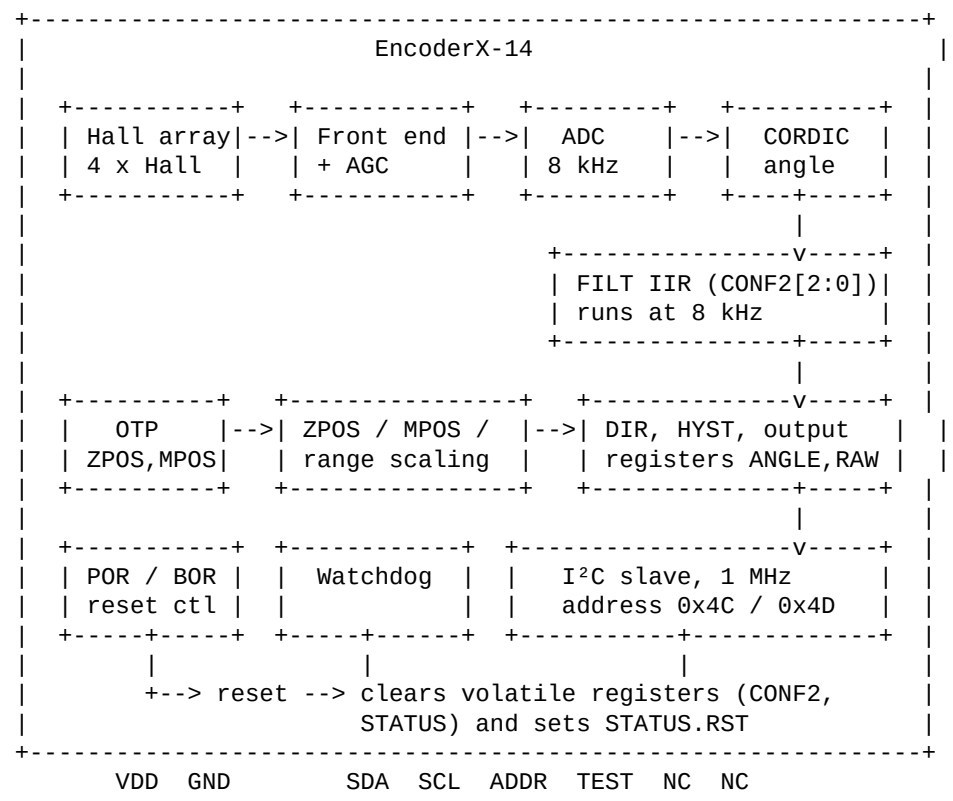


Figure 6-1. Functional block diagram.

6.1 Data path

The four Hall elements sense the two orthogonal field components of the rotating magnet. The front end amplifies them with a gain set by the automatic gain control (AGC, page 35). The ADC samples both channels at 8 kHz. The CORDIC engine converts each sample pair into an angle.

The FILT filter (page 31) is applied to the angle stream at the full 8 kHz rate, before the result is written to the RAW register. ZPOS, MPOS, DIR and HYST are applied after the filter, and the result goes to the ANGLE register.

6.2 Reset controller

Every reset source drives the same internal reset line. The reset line clears the volatile registers, including CONF2, regardless of the source. This is the root of errata E-3 (page 58).

7 Typical application: robot joint

Figure 7-1 shows the EncoderX-14 measuring the output angle of a robot elbow joint. The magnet is glued to the end of the joint shaft. The encoder sits on a small board on the joint housing, on the shaft axis.

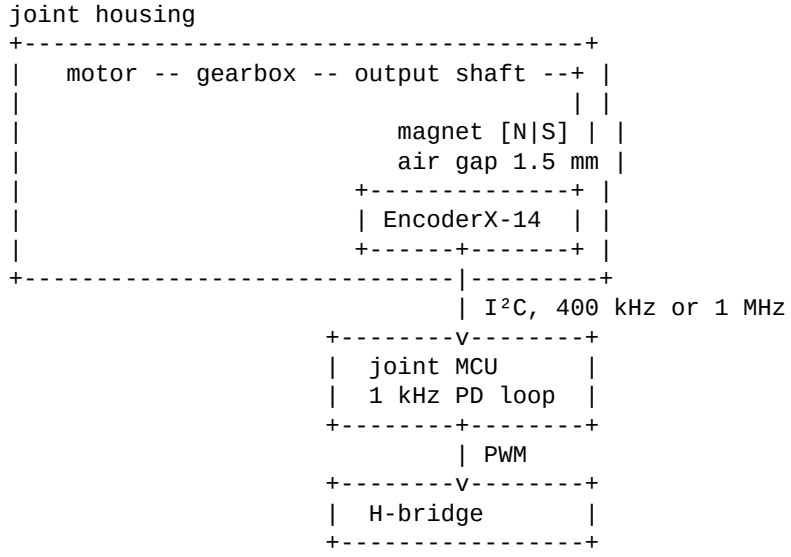


Figure 7-1. Robot joint with output-side angle sensing.

7.1 Control loop timing

The joint controller usually runs at 1 kHz. Each control period it reads the angle, computes the motor command and writes the PWM duty. The read takes a known time and the value returned is older than the moment of the read; see the latency table on page 12.

7.2 Derivative term

A PD or PID loop differentiates the measured angle. Differentiation amplifies high-frequency noise. With the FILT filter off, the raw output noise of 12 LSB RMS becomes a large velocity noise at a 1 kHz loop rate (page 48). This noise drives the motor, heats it, and can excite the mechanical resonance of a light arm. Page 49 explains how to choose a FILT setting for this application.

7.3 Brown-out in a joint

Joint boards often share a supply with the motor driver. A motor current spike can pull the 3.3 V rail below VBOR (2.6 V) for a few microseconds. That causes a brown-out reset, which clears CONF2 (page 58). The joint keeps running, but with the filter off. Page 53 describes how to detect this.

8 Ordering information

Part number	Package	Temperature	Packing
ENCX14-S8-T	SOIC-8	-40 to +125 °C	Tape and reel, 2500
ENCX14-S8-B	SOIC-8	-40 to +125 °C	Tube, 100
ENCX14-EVB	Board	0 to +70 °C	Evaluation board

8.1 Device marking

Line 1: X14
Line 2: YWW (year digit, week number)
Line 3: Lot code, 4 characters

8.2 Silicon revisions

REV register (0x01)	Marking suffix	Notes
0xA1	none	First production silicon
0xA2	"B"	E-1 and E-2 corrected

Errata E-3, E-4 and E-5 apply to both revisions. See page 57.

8.3 Evaluation board

The ENCX14-EVB carries the encoder, a magnet holder with a 6 mm x 2.5 mm NdFeB magnet, a 3.3 V regulator, I²C pull-ups (2.2 kOhm) and a 2.54 mm header for a host. Jumper J1 selects the I²C address. Jumper J2 inserts a 10 Ohm resistor in series with VDD so that brown-out behaviour can be reproduced with a bench supply.

8.4 Related documents

- ENCX14-AN01 Magnet and air gap selection
- ENCX14-AN02 Servo loop design with the EncoderX-14
- ENCX14-ER Errata sheet (the content is included on pages 57-59)

All related documents are fictional and listed for completeness only.

9 Absolute maximum ratings

Stresses beyond the values in table 9-1 may cause permanent damage. Operation at these values is not implied. Exposure to absolute maximum conditions for long periods may affect reliability.

Table 9-1. Absolute maximum ratings.

Parameter	Min	Max	Unit
Supply voltage VDD	-0.3	4.0	V
Voltage on SDA, SCL, ADDR	-0.3	VDD+0.3	V
Voltage on TEST	-0.3	VDD+0.3	V
Input current, any pin	-10	10	mA
Storage temperature	-55	150	°C
Junction temperature		150	°C
Magnetic flux density		unlimited	
ESD, human body model (HBM)		2000	V
ESD, charged device model (CDM)		500	V
Moisture sensitivity level		MSL 3	

9.1 Notes

1. The device survives any static magnetic field. The angle output is only valid between 30 mT and 90 mT (page 20).
2. TEST must be tied to GND in the application. Leaving it open can enter a factory test mode in which the I²C interface does not respond.
3. Exceeding 4.0 V on VDD can damage the OTP memory. OTP contents are not guaranteed after an overvoltage event.
4. A supply ramp faster than 1 V/µs can trigger more than one power-on reset. This is harmless; the last reset wins.

10 Recommended operating conditions

Table 10-1. Recommended operating conditions.

Parameter	Min	Typ	Max	Unit
Supply voltage VDD	3.0	3.3	3.6	V
Supply current, active		6.5	9.0	mA
Supply current, low-power mode		1.2	1.8	mA
Operating ambient temperature	-40		125	°C
Magnetic flux density	30	60	90	mT
Air gap (6 mm x 2.5 mm magnet)	0.5	1.5	3.0	mm
Rotational speed		0	30000	rpm
I²C clock frequency	0		1000	kHz
Supply ripple, 1 kHz to 1 MHz			50	mV pp

10.1 Supply decoupling

Place a 100 nF ceramic capacitor within 2 mm of the VDD pin and a 4.7 µF capacitor within 10 mm. The device draws current pulses at the 8 kHz sampling rate. Without the 4.7 µF capacitor these pulses can modulate the supply enough to add 1-2 LSB of noise.

10.2 Shared supplies

When the encoder shares a 3.3 V rail with a motor driver or a radio, consider a separate LDO or an RC filter (10 Ohm, 10 µF) on VDD. Supply dips below VBOR = 2.6 V cause a brown-out reset (page 11). The brown-out reset clears CONF2 (page 58).

10.3 Rotational speed

The angle output is valid up to 30,000 rpm. At high speed the output lags the true angle by the internal processing delay plus the FILT time constant (page 12 and page 31). The lag is a pure function of speed and can be compensated by the host if needed.

11 Supply, power-on reset and brown-out reset

11.1 Reset sources

The device has four reset sources. All four drive the same internal reset line and have the same effect on the registers.

Source	Trigger	Page
Power-on	VDD rises through VPOR = 2.8 V	11
Brown-out	VDD falls below VBOR = 2.6 V for > 2 µs	11
Watchdog	internal oscillator stalls for > 1 ms	54
Soft reset	host writes 0xA5 to CMD (0x30)	36

11.2 Effect of a reset

On every reset:

- CONF2 returns to 0x00. FILT = 0 (filter off), DIR = 0, HYST = 0, WD = 0.
- STATUS is cleared and then STATUS.RST (bit 0) is set.
- ZPOS, MPOS and CONF1 are reloaded from OTP.
- The I²C interface is idle and ready 1.5 ms after the reset ends.
- The first valid angle is available 2.0 ms after the reset ends.

11.3 Thresholds

Table 11-1. Power-on and brown-out thresholds.

Parameter	Symbol	Min	Typ	Max	Unit
Power-on reset threshold	VPOR	2.70	2.80	2.90	V
Brown-out reset threshold	VBOR	2.50	2.60	2.70	V
Brown-out filter time	tBOR	1	2	4	µs
Reset to I²C ready	tRDY		1.5	2.0	ms
Reset to first valid angle	tVAL		2.0	2.5	ms

11.4 Important: brown-out resets are silent

A brown-out reset can be shorter than one host control period. The host does not see a bus error and the angle keeps arriving after 2 ms. The only evidence is STATUS.RST, which stays set until the host reads STATUS (page 33). The configuration written by the host at start-up is lost: CONF2 is back to 0x00 and the noise filter is off. Errata E-3 on page 58 describes the consequences and the required workaround.

12 Timing, read latency and phase lag

12.1 Internal timing

Parameter	Typ	Max	Unit
Internal sampling period (8 kHz)	125	125	µs
CORDIC processing delay	60	80	µs
Register update period (ANGLE, RAW)	125	125	µs

12.2 Read latency seen by a host

The angle that a host receives is older than the moment the read completes. The age is the sum of the internal processing delay, the time since the last register update, the I²C transfer time and the time the host firmware takes to service the transfer.

For the typical case of a host running a 1 kHz control loop that starts one 3-byte burst read (page 40) per control period at 400 kHz, and that uses the value in the next control step, the effective latency is two control periods.

The I²C read latency is 2 ms for a host that polls once per 1 ms control period and uses the result in the following period.

Model the encoder in simulation as a pure delay of 2 ms on the true angle, followed by the FILT filter and the output noise.

12.3 Phase lag

A pure delay T adds a phase lag of 360° x f x T at frequency f. The FILT filter adds atan(2 x pi x f x tau). Table 12-1 lists the total phase lag for the 2 ms latency case.

Table 12-1. Phase lag from a 2 ms delay plus the FILT time constant.

Frequency	FILT=0	FILT=3 (1 ms)	FILT=5 (4 ms)	FILT=7 (16 ms)
1 Hz	0.7°	1.1°	2.2°	6.5°
4 Hz	2.9°	4.3°	8.6°	24.8°
10 Hz	7.2°	10.8°	21.3°	52.4°
30 Hz	21.6°	32.3°	58.6°	93.2°

A joint loop with a crossover between 3 and 10 Hz tolerates FILT = 3 with almost no extra phase lag. FILT = 7 costs about 25° at 4 Hz and can make a loop that was stable with FILT = 0 oscillate. See page 49.

13 I²C DC characteristics and address selection

Table 13-1. I²C DC characteristics (VDD = 3.0 to 3.6 V).

Parameter	Min	Max	Unit	
Input low voltage VIL	-0.3	0.3 VDD	V	
Input high voltage VIH	0.7 VDD	VDD+0.3	V	
Input hysteresis	0.05 VDD		V	
Output low voltage, 3 mA sink		0.4	V	
Output low voltage, 20 mA sink		0.4	V	(Fm+)
Input leakage current	-1	1	µA	
Pin capacitance		10	pF	

13.1 Pull-up resistors

SDA and SCL are open-drain. Choose pull-ups for the bus capacitance and speed:

Mode	Max clock	Typical pull-up	Max bus capacitance
Standard mode	100 kHz	4.7 kOhm	400 pF
Fast mode	400 kHz	2.2 kOhm	400 pF
Fast-mode plus	1 MHz	1.0 kOhm	550 pF

13.2 Address selection

The 7-bit I²C address is set by the ADDR pin at reset:

ADDR pin	7-bit address	Write byte	Read byte
GND	0x4C	0x98	0x99
VDD	0x4D	0x9A	0x9B

The ADDR pin is sampled on every reset, including brown-out resets. Do not drive ADDR from a GPIO that may change state while the device is running: a brown-out would then move the device to the other address.

13.3 Bus voltage

The I²C pins are not 5 V tolerant. A 5 V host needs a level shifter.

14 I²C AC timing

Table 14-1. I²C AC timing.

Parameter	Symbol	Fm (400 kHz)		Fm+ (1 MHz)		Unit
		Min	Max	Min	Max	
SCL clock frequency	fSCL	0	400	0	1000	kHz
Hold time START	tHD;STA	0.6		0.26		µs
SCL low period	tLOW	1.3		0.5		µs
SCL high period	tHIGH	0.6		0.26		µs
Setup time repeated START	tSU;STA	0.6		0.26		µs
Data hold time	tHD;DAT	0		0		µs
Data setup time	tSU;DAT	100		50		ns
Rise time SDA, SCL	tr		300		120	ns
Fall time SDA, SCL	tf		300		120	ns
Setup time STOP	tSU;STO	0.6		0.26		µs
Bus free time	tBUF	1.3		0.5		µs

14.1 Transfer durations

A 3-byte burst read from STATUS (page 40) consists of the write address byte, the register pointer byte, a repeated START, the read address byte and 3 data bytes: 6 bytes with acknowledgements, 54 clock cycles plus the START, repeated START and STOP conditions.

Bus speed	Burst read time
100 kHz	540 µs
400 kHz	135 µs
1 MHz	54 µs

14.2 Clock stretching

The device never stretches the clock. The host may use any clock frequency up to the limit of the selected mode.

14.3 General call

The device ignores the general call address (0x00).

15 Pinout

SOIC-8, top view

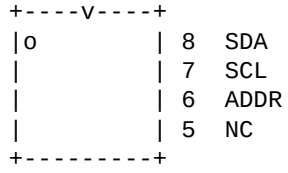


Table 15-1. Pin functions.

Pin	Name	Type	Function
1	VDD	Supply	3.3 V supply. Decouple with 100 nF + 4.7 µF.
2	GND	Supply	Ground.
3	TEST	Input	Factory test. Tie to GND.
4	NC	-	Not connected. Leave open or tie to GND.
5	NC	-	Not connected. Leave open or tie to GND.
6	ADDR	Input	I²C address select, sampled at reset (page 13).
7	SCL	Input	I²C clock, open-drain bus.
8	SDA	I/O	I²C data, open-drain.

15.1 Sensor centre

The centre of the Hall array is in the centre of the package body, within ±0.1 mm. Use the package outline, not the pin row, to place the magnet axis (page 21).

15.2 Direction of rotation

Looking at the top of the package, with DIR = 0 (default), the angle code increases when the magnet rotates clockwise. Setting CONF2.DIR = 1 reverses the direction (page 32).

Important: because DIR lives in CONF2, a brown-out reset also returns DIR to 0 (page 58). A host that relies on DIR = 1 will see the direction reverse after a brown-out. In a position loop this turns negative feedback into positive feedback. Prefer to handle direction in host software, or rewrite CONF2 after every reset (page 45).

16 Package dimensions (SOIC-8)

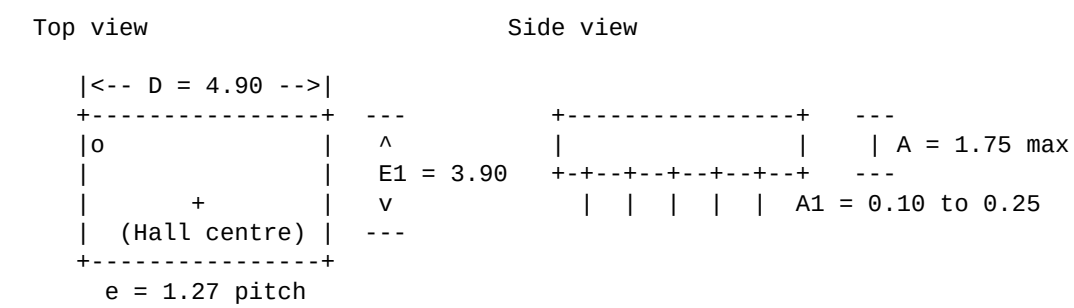


Table 16-1. SOIC-8 dimensions (mm).

Symbol	Description	Min	Nom	Max
A	Total height			1.75
A1	Standoff	0.10		0.25
b	Lead width	0.31		0.51
D	Body length	4.80	4.90	5.00
E	Width across leads	5.80	6.00	6.20
E1	Body width	3.80	3.90	4.00
e	Lead pitch		1.27	
L	Foot length	0.40		1.27
dZ	Hall array depth below top surface		0.30	

16.1 Hall array position

The Hall array is centred in the body in both x and y within ±0.10 mm, and sits 0.30 mm below the top surface of the mould compound. Add 0.30 mm to the mechanical air gap to obtain the magnetic air gap.

16.2 Recommended land pattern

Pad width 0.60 mm, pad length 1.55 mm, pad row pitch 5.40 mm, pad pitch 1.27 mm.

Keep the area under the package free of copper pours on the top layer to avoid eddy-current effects at high rotational speed.

17 PCB layout guidance

17.1 Placement

- Place the decoupling capacitors on the same layer as the encoder, as close to pins 1 and 2 as possible.
- Route SDA and SCL as a pair, away from motor phase traces.
- Keep current-carrying traces at least 5 mm from the package. A trace carrying 1 A at 5 mm produces about 40 µT, which is small compared with the 30 mT minimum signal, but motor phase currents of 10 A or more are not.

17.2 Ground

Connect GND to a solid ground plane. Do not route the motor return current under the encoder.

17.3 Supply

If the board carries a motor driver, give the encoder its own supply branch from the regulator, with an RC filter (10 Ohm, 10 µF). This reduces the probability of a brown-out reset when the motor starts (page 53). It does not remove that probability, so firmware must still handle STATUS.RST (page 45).

17.4 Connectors and cables

When the encoder board is connected to the host by a cable longer than 30 cm, use fast mode (400 kHz) rather than fast-mode plus, add 100 Ohm series resistors on SDA and SCL at the encoder side, and keep VDD and GND in the same cable as the bus. Long cables raise the chance of supply dips at the encoder.

17.5 Magnet keep-out

Keep ferromagnetic parts (steel screws, shields) at least 5 mm away from the magnet. Ferromagnetic parts distort the field and add an angle error that repeats once per revolution (page 27).

18 Soldering and handling

18.1 Reflow profile

The device is qualified for lead-free reflow according to the following profile:

Stage	Value
Preheat, 150 to 200 °C	60 to 120 s
Time above 217 °C	60 to 150 s
Peak temperature	260 °C, max 10 s within 5 °C of peak
Ramp-up rate	3 °C/s max
Ramp-down rate	6 °C/s max
Maximum number of reflows	3

18.2 Moisture sensitivity

The device is MSL 3. After opening the dry pack, mount within 168 hours at 30 °C and 60 % RH or less, or bake at 125 °C for 24 hours.

18.3 Hand soldering

Hand soldering is allowed with a soldering iron at up to 350 °C for 3 s per lead. Do not touch the top of the package with the iron; the mould compound above the Hall array is thin.

18.4 Cleaning

Use no-clean flux where possible. If cleaning is needed, use isopropyl alcohol. Do not use ultrasonic cleaning; it can crack the die attach.

18.5 Handling the magnet

NdFeB magnets are brittle and attract each other strongly. Store magnets apart from encoders and from each other. Do not machine NdFeB magnets.

18.6 OTP and soldering

OTP contents are not affected by reflow. Program OTP after assembly (page 46), because the zero position depends on the final mechanical mounting.

19 Magnet selection

19.1 Magnet type

Use a diametrically magnetised cylindrical magnet. The north and south poles are on opposite halves of the cylinder, not on its end faces. An axially magnetised magnet does not work with this device.

19.2 Recommended magnets

Table 19-1. Recommended magnets.

Material	Diameter	Height	Air gap range	Notes
NdFeB N35	6 mm	2.5 mm	0.5 to 3.0 mm	Reference magnet
NdFeB N35	8 mm	3.0 mm	1.0 to 4.0 mm	Larger air gap
NdFeB N35	4 mm	2.0 mm	0.5 to 1.8 mm	Small joints
SmCo 2:17	6 mm	2.5 mm	0.5 to 2.4 mm	High temperature

19.3 Temperature

The remanence of NdFeB falls by about 0.12 % per °C. Between 25 °C and 125 °C the field falls by about 12 %. The AGC (page 35) compensates for the loss automatically as long as the field stays above 30 mT. Check the field at the highest operating temperature and largest air gap.

19.4 Magnet holder

Fix the magnet in a non-ferromagnetic holder (brass, aluminium, plastic). Use an adhesive that keeps the magnet centred to within 0.1 mm. A magnet that shifts in its holder causes a slowly growing angle error.

19.5 Magnet and noise

Output noise depends on the field. At 30 mT the raw noise is about 20 % higher than at 60 mT. Table 31-2 is measured at 60 mT. See page 26.

20 Air gap and field strength

20.1 Field versus air gap

Table 20-1 gives the field at the Hall array for the reference magnet (NdFeB N35, 6 mm x 2.5 mm) as a function of the mechanical air gap tAIR, measured from the top of the package to the face of the magnet.

Table 20-1. Field versus air gap, reference magnet, 25 °C.

tAIR (mm)	Field (mT)	AGC (typ)	Status flags
0.25	121	0	MAGH
0.50	98	12	MAGH (marginal)
0.75	88	24	-
1.00	79	41	-
1.50	62	78	-
2.00	48	112	-
2.50	37	151	-
3.00	30	186	MAGL (marginal)
3.50	24	220	MAGL
4.00	19	255	MAGL

Rev 1.1 corrected the tAIR maximum from 3.5 mm to 3.0 mm.

20.2 Choosing the air gap

Choose a nominal air gap in the middle of the valid field range so that tolerances and temperature drift keep the field within 30 to 90 mT. For the reference magnet, 1.5 mm is the recommended nominal value.

20.3 Field and noise

The output noise in table 31-2 is specified at 60 mT. At 30 mT add 20 %. At 90 mT subtract 5 %. These factors apply to all FILT settings.

20.4 Axial play

Axial play of the shaft changes the air gap. A play of ±0.2 mm around a 1.5 mm nominal gap changes the field by about ±7 mT, which is well within the AGC range and has no effect on the angle.

21 Mechanical alignment

21.1 Lateral displacement

The magnet axis should coincide with the centre of the Hall array (page 16). Lateral displacement causes an angle error that repeats once per revolution.

Lateral displacement	Peak angle error (reference magnet, 1.5 mm gap)
0.10 mm	0.10°
0.25 mm	0.28°
0.50 mm	0.62°
1.00 mm	1.50°

21.2 Tilt

Tilt of the magnet relative to the package surface causes an angle error that repeats twice per revolution.

Tilt	Peak angle error
1°	0.05°
2°	0.12°
5°	0.35°

21.3 Run-out

Shaft run-out moves the magnet in a circle. The error is the same as a lateral displacement equal to the run-out radius, but its phase rotates with the shaft.

21.4 Alignment procedure

1. Mount the encoder board and the magnet holder.
2. Rotate the shaft slowly through one revolution while reading ANGLE.
3. Compare ANGLE with a reference (a second encoder or an optical index).
4. Fit a sine at once and twice the rotation frequency to the difference.
5. If the once-per-revolution amplitude exceeds the budget, re-centre the board.

21.5 Effect on servo loops

Alignment errors are smooth and repeat with the shaft angle. They appear as a slow position error, not as noise. They are not reduced by FILT (page 31).

22 Alignment error and its effect on accuracy

22.1 Error budget example

For a robot elbow joint with the reference magnet at a 1.5 mm gap:

Source	Peak error
Lateral displacement 0.25 mm	0.28°
Tilt 1°	0.05°
Integral non-linearity (page 27)	0.10°
Stray field from motor (5 mT)	0.15°
Total (root sum of squares)	0.35°

At the end of a 300 mm forearm, 0.35° corresponds to 1.8 mm. This is a slow error that depends on the joint angle. It does not cause oscillation.

22.2 Noise is a separate budget

Output noise is random and broadband (page 26). With FILT = 0 the noise is 12 LSB RMS = 0.26° RMS. At the end of a 300 mm forearm this is 1.4 mm RMS of apparent position noise. A position loop with a high proportional gain and a derivative term turns this noise into motor torque, which can be seen as vibration of the arm. The noise budget is set by FILT, not by alignment.

22.3 Calibration

Alignment errors can be calibrated out with a look-up table in the host firmware. Noise cannot be calibrated out; it can only be filtered (page 31, page 49).

22.4 Summary

Error type	Varies with	Reduced by
Alignment	shaft angle	mechanical fix, calibration
Stray field	motor current	shielding, distance
Non-linearity	shaft angle	calibration
Output noise	nothing (random)	FILT[2:0], host filtering

23 Signal chain overview

The signal chain converts the magnetic field into a 14-bit angle in five stages:

1. Hall array. Four Hall plates measure the two field components Bx, By.
2. Front end. Differential amplifiers with automatic gain control.
3. ADC. Two 12-bit sigma-delta converters sampled at 8 kHz.
4. CORDIC. Computes atan2(By, Bx) to 14 bits once per sample.
5. FILT. A first-order IIR low-pass filter on the angle (page 31).

After the filter, the output stage applies ZPOS, MPOS, DIR and HYST and updates ANGLE and RAW once every 125 µs.

23.1 Angle wrap handling in the filter

The filter works on the unwrapped angle, so a transition from code 16383 to code 0 does not produce a filtered spike. The output is wrapped again to 14 bits after the filter.

23.2 Where the noise comes from

Most of the output noise comes from the Hall plates and the front end (thermal and 1/f noise) and from ADC quantisation. CORDIC rounding adds less than 0.3 LSB RMS. Page 26 gives the breakdown.

23.3 Where the delay comes from

Stage	Delay
ADC conversion	40 µs
CORDIC	20 µs
FILT (when enabled)	tau = 2^n / 8000 s, see table 31-2
Register update	0 to 125 µs (sampling phase)
I²C transfer	see page 14
Host scheduling	depends on the host (page 12)

The total delay seen by a typical 1 kHz host is 2 ms (page 12), plus the FILT time constant if the filter is enabled.

24 Hall array and front end

24.1 Hall array

Four Hall plates are arranged on a circle of 1.0 mm radius at 0°, 90°, 180° and 270°. Opposite plates are subtracted to cancel any uniform external field along the package normal. The difference signals are proportional to Bx and By of the magnet field.

24.2 Common-mode field rejection

A uniform external field (for example the field of a nearby motor) is rejected by the differential arrangement with a ratio of about 40 dB. A non-uniform stray field is not rejected and appears as an angle error (page 22).

24.3 Automatic gain control

The AGC adjusts the front-end gain so that the amplitude of the Bx/By signals at the ADC input is constant. The AGC value (0 to 255) is readable in register AGC (0x1A, page 35). A low AGC value means a strong field; a high value means a weak field.

The AGC loop time constant is 10 ms. The AGC does not affect the angle directly, but a fast change of air gap (axial shock) can cause a short angle disturbance while the AGC settles.

24.4 Front-end noise

The input-referred noise of the front end is 2.1 µT RMS in the 4 kHz bandwidth of the ADC. Together with the Hall plate noise and quantisation, this gives the 12 LSB RMS total output noise at FILT = 0 (page 26).

24.5 Front-end saturation

Above 120 mT the front end saturates. MAGH is set. The angle is invalid while MAGH is set and must not be used for control.

25 ADC and CORDIC angle computation

25.1 ADC

Two sigma-delta ADCs convert Bx and By simultaneously at 8 kHz. Each conversion has 12 effective bits. Simultaneous sampling avoids an angle error that would otherwise grow with rotational speed.

25.2 CORDIC

The CORDIC engine computes the angle atan2(By, Bx) in 16 iterations and rounds the result to 14 bits. The CORDIC result is available 60 µs after the start of the ADC sample (page 23).

25.3 Output resolution versus noise

The output has 14 bits (0.022° per LSB). With FILT = 0 the output noise is 12 LSB RMS, so only about 7.7 bits are noise-free in a single read. The FILT filter trades bandwidth for noise (page 31):

FILT	Noise (LSB RMS)	Noise-free bits (approx.)
0	12.0	7.7
3	3.10	9.6
5	1.51	10.7
7	0.75	11.7

Noise-free bits are computed as 14 - log2(6.6 x noise), where 6.6 x RMS is the peak-to-peak noise for a Gaussian distribution at 99.9 % coverage.

25.4 Linearity

The CORDIC engine is exact to within 0.25 LSB. Non-linearity is dominated by magnetic effects (page 27).

25.5 Output data rate

ANGLE and RAW are updated every 125 µs (8 kHz). A host that reads faster than 8 kHz receives repeated values.

26 Noise sources

26.1 Noise breakdown at FILT = 0, 60 mT, 25 °C

Source	Contribution (LSB RMS)
Hall plate thermal noise	9.8
Front-end amplifier noise	6.2
ADC quantisation	2.4
Supply-coupled noise (with 4.7 µF)	1.1
CORDIC rounding	0.3
Total (root sum of squares)	12.0

26.2 Spectrum

With FILT = 0 the noise is white from DC to 4 kHz (half the 8 kHz sampling rate). A host that samples the angle at 1 kHz sees the whole 0 to 4 kHz noise power aliased into its 0 to 500 Hz band. The RMS value per read is therefore still 12 LSB.

26.3 Effect of FILT

FILT is a first-order low-pass filter running at 8 kHz. It reduces the bandwidth of the white noise and therefore its RMS value. The noise gain of the filter is sqrt(alpha / (2 - alpha)), where alpha = 2^-n (page 31).

26.4 Dependence on field

Field	Noise multiplier
30 mT	1.20
45 mT	1.08
60 mT	1.00
90 mT	0.95

26.5 Dependence on temperature

The noise increases by about 0.1 % per °C above 25 °C, so by about 10 % at 125 °C.

26.6 Noise is not drift

Noise as specified here is the fast, random part of the error. Slow variations with temperature or shaft angle are covered on page 27.

27 Accuracy and linearity

27.1 Definitions

INL	Integral non-linearity: the largest deviation of the output from a straight line through the full revolution, after removing offset.
Drift	Change of the output at a fixed angle over temperature.
Noise	Random variation of consecutive reads at a fixed angle (page 26).

27.2 Specifications

Table 27-1. Accuracy (reference magnet, 1.5 mm air gap, perfect alignment).

Parameter	Typ	Max	Unit
INL, 25 °C	0.10	0.25	°
INL, -40 to 125 °C	0.15	0.40	°
Zero drift, -40 to 125 °C	0.05	0.12	°
Hysteresis (HYST = 0)	0	0	LSB
Output noise, FILT = 0	12.0	14.0	LSB RMS
Output noise, FILT = 3	3.10	3.6	LSB RMS
Output noise, FILT = 7	0.75	0.9	LSB RMS

27.3 Notes

1. INL includes the CORDIC error and the effect of magnet inhomogeneity. It excludes alignment errors (page 21).
2. Noise values are at 60 mT. See page 26 for other field values.
3. Max noise values are the 3-sigma production limits.

27.4 Improving accuracy

- Calibrate the INL with a look-up table in the host.
- Use FILT to reduce noise, at the cost of phase lag (page 12).
- Average several reads in the host only if the host reads faster than the ANGLE update rate; otherwise averaging repeated values does nothing (page 25).

28 Register map summary

Table 28-1. Register map.

Addr	Name	Access	Default	Description	Page
0x00	CHIP_ID	R	0x14	Device identifier	29
0x01	REV	R	0xA1/A2	Silicon revision	29
0x02	CONF1	RW/OTP	0x00	Power mode, ADC options	29
0x03	ZPOS_H	RW/OTP	0x00	Zero position, bits 13..8	30
0x04	ZPOS_L	RW/OTP	0x00	Zero position, bits 7..0	30
0x05	MPOS_H	RW/OTP	0x3F	Maximum position, bits 13..8	30
0x06	MPOS_L	RW/OTP	0xFF	Maximum position, bits 7..0	30
0x07	-	-	-	Reserved, do not write	-
0x08	CONF2	RW	0x00	FILT, DIR, HYST, WD (volatile)	31, 32
0x09	-	-	-	Reserved, do not write	-
0x0A	-	-	-	Reserved, do not write	-
0x0B	STATUS	RC	0x01	RST, MAGL, MAGH, MD, OVF	33
0x0C	ANGLE_H	R	-	Scaled angle, bits 13..8	34
0x0D	ANGLE_L	R	-	Scaled angle, bits 7..0	34
0x0E	RAW_H	R	-	Raw angle, bits 13..8	34
0x0F	RAW_L	R	-	Raw angle, bits 7..0	34
0x10	-	-	-	Reserved (see errata E-4)	59
0x1A	AGC	R	-	Automatic gain control value	35
0x1B	MAG_H	R	-	Field magnitude, bits 11..8	35
0x1C	MAG_L	R	-	Field magnitude, bits 7..0	35
0x30	CMD	W	-	Soft reset, OTP burn	36

28.1 Volatile and non-volatile registers

CONF1, ZPOS and MPOS have OTP shadows: after a reset they are reloaded from OTP. CONF2 has no OTP shadow and returns to 0x00 after every reset. STATUS returns to 0x01 (RST set) after every reset.

28.2 Reserved addresses

Writes to reserved addresses are ignored. Reads return 0x00, except 0x10 (errata E-4, page 59).

29 CHIP_ID, REV and CONF1

29.1 CHIP_ID (0x00, R)

Bits	Field	Value
7:0	CHIP_ID	0x14

Read CHIP_ID at start-up to confirm that the device answers on the expected address. A value other than 0x14 means a wrong address, a bus fault, or a different device.

29.2 REV (0x01, R)

Bits	Field	Value
7:4	MAJOR	0xA
3:0	MINOR	0x1 (first silicon) or 0x2 (E-1, E-2 corrected)

29.3 CONF1 (0x02, RW, OTP shadow)

Bits	Field	Default	Description
7:6	-	0	Reserved, write 0
5:4	PM[1:0]	0	Power mode: 0 = active, 1 = low power 1, 2 = low power 2, 3 = reserved
3	ADCSLOW	0	1 = halve the ADC rate to 4 kHz (saves 1 mA, doubles FILT time constants)
2:0	-	0	Reserved, write 0

CONF1 is reloaded from OTP after every reset. If CONF1 has not been burnt, its OTP value is 0x00.

29.4 Low-power modes

PM	Sampling	Current	Note
0	continuous 8 kHz	6.5 mA	Use for servo loops
1	8 kHz, 50 % duty	3.4 mA	Adds 125 µs of jitter
2	8 kHz, 12.5 % duty	1.2 mA	Not for servo loops

Entering low-power mode takes longer than specified on rev 0xA1 silicon; see errata E-5 on page 59.

30 ZPOS and MPOS (zero and maximum position)

30.1 ZPOS (0x03-0x04, RW, OTP shadow)

Register	Bits	Field	Description
ZPOS_H	5:0	ZPOS[13:8]	Zero position, high bits
ZPOS_L	7:0	ZPOS[7:0]	Zero position, low bits

ZPOS is subtracted from the filtered angle before it is written to ANGLE. Write ZPOS_H first, then ZPOS_L. The new zero takes effect after the write to ZPOS_L.

30.2 MPOS (0x05-0x06, RW, OTP shadow)

Register	Bits	Field	Description
MPOS_H	5:0	MPOS[13:8]	Maximum position, high bits
MPOS_L	7:0	MPOS[7:0]	Maximum position, low bits

When MPOS is less than 0x3FFF, ANGLE is scaled so that the range from ZPOS to MPOS spans the full 14-bit output. The default 0x3FFF gives a full 360° range with no scaling.

30.3 Scaling and noise

Scaling a small range to the full output also scales the noise. For a 90° range (MPOS = 4096) the noise in ANGLE is four times the noise in RAW, expressed in output codes. Noise expressed in degrees is unchanged.

30.4 Programming ZPOS and MPOS permanently

ZPOS and MPOS written over I²C are volatile until burnt into OTP. A reset reloads them from OTP. To make them permanent, use the OTP burn command on page 36. OTP can be burnt only once.

30.5 Host-side alternative

Many hosts keep ZPOS = 0 and MPOS = 0x3FFF and apply the offset in firmware. This avoids OTP programming in production and avoids surprises after a reset.

31 CONF2 and the FILT[2:0] filter

31.1 CONF2 (0x08, RW, volatile)

Table 31-1. CONF2 bit fields.

Bits	Field	Default	Description
7	-	0	Reserved, write 0
6	WD	0	Watchdog enable (page 32, page 54)
5:4	HYST[1:0]	0	Output hysteresis, 0 to 3 LSB (page 32)
3	DIR	0	Direction: 0 = clockwise increases (page 32)
2:0	FILT[2:0]	0	Output noise filter setting n, 0 = off

CONF2 has no OTP shadow. Its value after any reset is 0x00, so FILT = 0 and the filter is off. This includes brown-out resets (errata E-3, page 58).

31.2 The FILT filter

FILT selects a first-order IIR low-pass filter that runs on the angle at the 8 kHz internal sample rate, before the RAW and ANGLE registers:

    y[k] = y[k-1] + alpha x (x[k] - y[k-1]),   alpha = 2^-n, n = FILT[2:0]

With n = 0 the filter is bypassed. The time constant is tau = 2^n / 8000 s and the -3 dB bandwidth is approximately 1 / (2 x pi x tau). For white input noise the output noise is multiplied by sqrt(alpha / (2 - alpha)).

Table 31-2. Output noise and bandwidth versus FILT (60 mT, 25 °C).

FILT n	alpha	tau	Noise (LSB RMS)	Noise (° RMS)	Approx. BW
0	1	off	12.0	0.264	-
1	1/2	0.25 ms	6.93	0.152	637 Hz
2	1/4	0.5 ms	4.54	0.100	318 Hz
3	1/8	1 ms	3.10	0.068	159 Hz
4	1/16	2 ms	2.155	0.047	79.6 Hz
5	1/32	4 ms	1.51	0.033	39.8 Hz
6	1/64	8 ms	1.06	0.023	19.9 Hz
7	1/128	16 ms	0.75	0.016	9.9 Hz

31.3 Writing FILT

To set FILT = 3 with the other CONF2 fields at their defaults, write 0x03 to address 0x08. To change FILT without changing other fields, read CONF2, clear bits 2..0, OR in the new value and write it back.

31.4 Important

- FILT = 0 after every reset. A host that needs the filter must write CONF2 at start-up and again after every reset (page 45, page 58).
- FILT adds phase lag (table 12-1). Do not choose a time constant that is large compared with the period of the fastest motion you control.
- If ADCSLOW = 1 (page 29), the filter runs at 4 kHz and every tau in table 31-2 doubles.

32 CONF2 other fields (DIR, HYST, WD)

32.1 DIR (CONF2 bit 3)

DIR = 0	angle increases for clockwise rotation, seen from the top
DIR = 1	angle increases for counter-clockwise rotation

DIR is volatile. After a reset it returns to 0. A host that sets DIR = 1 must set it again after every reset (page 15, page 58).

32.2 HYST[1:0] (CONF2 bits 5..4)

HYST	Hysteresis on ANGLE
0	none
1	1 LSB
2	2 LSB
3	3 LSB

Hysteresis suppresses toggling of the last bits when the shaft is at rest. It is applied after FILT and does not reduce noise in a moving shaft.

Important: hysteresis is not a noise filter for servo loops. With HYST > 0 the output moves in steps, which a derivative term turns into spikes. Use FILT instead (page 49). Hysteresis does not affect RAW.

32.3 WD (CONF2 bit 6)

WD = 0	watchdog disabled (default)
WD = 1	watchdog enabled; the device resets itself if the internal oscillator stalls for more than 1 ms (page 54)

Because the watchdog reset is also a reset, it clears CONF2, including WD itself. The host sees STATUS.RST = 1 after a watchdog reset (page 33).

32.4 Example values

Purpose	CONF2 value
Defaults (after any reset)	0x00
FILT = 3, everything else default	0x03
FILT = 3, DIR = 1	0x0B
FILT = 5, HYST = 1, WD = 1	0x55

33 STATUS register

33.1 STATUS (0x0B, RC)

Table 33-1. STATUS bit fields.

Bits	Field	Type	Description
7:5	-	R	Reserved, read as 0
4	OVF	RC	CORDIC overflow (field too weak to compute an angle)
3	MD	R	Magnet detected (field between 20 mT and 120 mT)
2	MAGH	R	Field too strong (> 90 mT)
1	MAGL	R	Field too weak (< 30 mT)
0	RST	RC	A reset has occurred since STATUS was last read

33.2 The RST flag is sticky

RST is set by every reset: power-on, brown-out, watchdog and soft reset (page 11). It stays set until the host reads STATUS. Reading STATUS clears RST and OVF. The value returned by that read still shows RST = 1; the flag is cleared after the read completes.

This means:

- The host sees RST = 1 exactly once per reset (or once for several resets that happen between two reads).
- The host must act on RST in the same control step in which it reads it. The next read returns RST = 0 even though the configuration is still lost.
- RST = 1 means CONF2 is 0x00 again: FILT is off, DIR = 0, HYST = 0, WD = 0 (errata E-3, page 58).

33.3 Reading STATUS together with the angle

The recommended way to read STATUS is the 3-byte burst read starting at 0x0B (page 40). It returns STATUS, ANGLE_H and ANGLE_L in one transaction, so the host checks RST on every control step at no extra cost.

33.4 MAGL, MAGH and MD

These flags reflect the current field. They are not sticky. The angle is not valid when MAGL or MAGH is set or MD is clear. See page 52.

34 ANGLE and RAW output registers

34.1 ANGLE (0x0C-0x0D, R)

Register	Bits	Field	Description
ANGLE_H	7:6	-	Read as 0
ANGLE_H	5:0	ANGLE[13:8]	Scaled angle, high bits
ANGLE_L	7:0	ANGLE[7:0]	Scaled angle, low bits

ANGLE is the filtered angle after ZPOS, MPOS, DIR and HYST.

34.2 RAW (0x0E-0x0F, R)

Register	Bits	Field	Description
RAW_H	7:6	-	Read as 0
RAW_H	5:0	RAW[13:8]	Raw angle, high bits
RAW_L	7:0	RAW[7:0]	Raw angle, low bits

RAW is the filtered angle before ZPOS, MPOS, DIR and HYST. FILT does act on RAW: RAW is taken after the filter (page 6).

34.3 Data consistency

A multi-byte read latches the high and low bytes together at the start of the read, so ANGLE_H and ANGLE_L (or RAW_H and RAW_L) always belong to the same sample when they are read in one transaction. Two separate single-byte reads can combine bytes from different samples.

34.4 Converting to radians

```
angle_rad = code x 2 x pi / 16384
angle_deg = code x 360 / 16384
```

One code is 383.5 µrad = 0.02197°.

34.5 Unwrapping

The output wraps from 16383 to 0. For a joint that moves less than half a revolution between two reads, unwrap by adding or subtracting 16384 when the difference between consecutive reads exceeds 8192 in magnitude.

35 AGC and MAG registers

35.1 AGC (0x1A, R)

Bits	Field	Description
7:0	AGC	Automatic gain control value, 0 (strong) to 255 (weak)

The AGC value is a convenient indicator of the air gap. Table 20-1 lists typical values for the reference magnet. A slowly rising AGC value over the life of a product indicates a growing air gap or a weakening magnet.

35.2 MAG (0x1B-0x1C, R)

Register	Bits	Field	Description
MAG_H	3:0	MAG[11:8]	Field magnitude after AGC, high bits
MAG_L	7:0	MAG[7:0]	Field magnitude after AGC, low bits

MAG is the magnitude sqrt(Bx^2 + By^2) after the AGC. With the AGC in its range, MAG is constant at about 2048. MAG that varies by more than ±10 % during one revolution indicates misalignment (page 21).

35.3 Using AGC and MAG in production test

Test	Pass criterion
AGC at nominal air gap	40 to 120 (reference magnet)
MAG variation per revolution	within ±10 %
STATUS.MD	1
STATUS.MAGL, STATUS.MAGH	0

35.4 AGC and noise

A high AGC value (weak field) means more front-end gain and more noise. The noise multipliers on page 26 apply. If FILT is chosen from table 31-2 for a 60 mT design, check the noise again at the weakest expected field.

36 CMD register and OTP

36.1 CMD (0x30, W)

Value	Command
0xA5	Soft reset
0x80	Burn OTP (CONF1, ZPOS, MPOS)
others	Ignored

36.2 Soft reset

Writing 0xA5 to CMD resets the device. The effect is identical to a power-on reset (page 11): CONF2 = 0x00, STATUS.RST = 1, CONF1, ZPOS and MPOS reloaded from OTP. The device does not acknowledge the data byte of the soft-reset write; a NACK on that byte is expected.

36.3 OTP burn

Writing 0x80 to CMD copies the current CONF1, ZPOS and MPOS register values into OTP. The burn takes 10 ms. Do not access the device during the burn.

Requirements:

- VDD between 3.2 V and 3.6 V during the burn (errata E-1, page 57).
- The burn can be done only once. A second burn command is ignored.
- CONF2 cannot be burnt. FILT, DIR, HYST and WD must always be written by the host after a reset.

36.4 Verifying the burn

1. Write 0xA5 to CMD (soft reset).
2. Wait 2 ms.
3. Read ZPOS, MPOS and CONF1 and compare them with the intended values.
4. Read STATUS and confirm RST = 1 (the soft reset was seen).

36.5 Why CONF2 is not in OTP

CONF2 controls behaviour that a host may need to change at run time (for example a stronger filter at standstill). Keeping CONF2 volatile allows this, at the cost that the host must rewrite CONF2 after every reset.

37 I²C protocol overview

37.1 Bus parameters

Address	0x4C (ADDR = GND) or 0x4D (ADDR = VDD), 7-bit
Speeds	100 kHz, 400 kHz, 1 MHz
Byte order	Most significant byte first
Pointer	8-bit register pointer, auto-increment on read and write

37.2 Transactions

Transaction	Sequence
Write register	S, addr+W, ptr, data..., P
Read register	S, addr+W, ptr, Sr, addr+R, data..., P
Read current	S, addr+R, data..., P (uses the last pointer)

S = START, Sr = repeated START, P = STOP.

37.3 Pointer behaviour

The register pointer persists between transactions. After a read of N bytes starting at pointer p, the pointer is p + N (with the wrap behaviour described on page 41). A "read current" transaction therefore continues where the last read ended, which is rarely what a control loop wants. Always set the pointer explicitly.

37.4 Reset and the bus

During a reset the device releases SDA and does not respond to its address. A transaction that starts during the 1.5 ms after a reset is NACKed. After a brown-out the pointer is reset to 0x00.

37.5 Typical control-loop traffic

One 3-byte burst read from 0x0B per control period (page 40). Configuration writes only at start-up and after STATUS.RST (page 45).

38 Write transaction

38.1 Sequence

```
  S | 0x98 (0x4C + W) | A | ptr | A | data0 | A | data1 | A | ... | P
```

Each data byte is written to the register at the pointer, and the pointer is incremented after every byte.

38.2 Example: enable the filter

Set FILT = 3 (CONF2 = 0x03):

```
  S | 0x98 | A | 0x08 | A | 0x03 | A | P
```

38.3 Example: set ZPOS to 0x1234

```
  S | 0x98 | A | 0x03 | A | 0x12 | A | 0x34 | A | P
```

38.4 Writes to read-only and reserved registers

Writes to read-only and reserved registers are acknowledged and ignored. The pointer is still incremented.

38.5 Write confirmation

The device does not report whether a write took effect. To confirm a configuration write, read the register back. A host that writes CONF2 and immediately reads it back can still lose the value later through a reset; read-back confirms the write, not that the value will stay (page 58).

38.6 Timing between writes

There is no minimum time between write transactions, except after a soft reset (2 ms) and during an OTP burn (10 ms).

39 Read transaction

39.1 Sequence

```
  S | 0x98 | A | ptr | A | Sr | 0x99 | A | data0 | A | data1 | A | ... | NA | P
```

The host acknowledges every byte except the last, which it NACKs before the STOP.

39.2 Example: read CHIP_ID

```
  S | 0x98 | A | 0x00 | A | Sr | 0x99 | A | 0x14 | NA | P
```

39.3 Example: read ANGLE

```
  S | 0x98 | A | 0x0C | A | Sr | 0x99 | A | ANGLE_H | A | ANGLE_L | NA | P
```

The two bytes are latched together (page 34).

39.4 Reading STATUS clears flags

Any read transaction that includes address 0x0B clears STATUS.RST and STATUS.OVF after the byte has been transmitted (page 33). This includes burst reads that start below 0x0B and run through it.

39.5 Read latency

The data returned are the values latched at the start of the read. The latency seen by a typical host is given on page 12.

40 Burst read of STATUS and ANGLE

40.1 The recommended per-cycle read

A 3-byte burst read starting at STATUS returns the status and the angle in one transaction:

```
S | 0x98 | A | 0x0B | A | Sr | 0x99 | A | STATUS | A | ANGLE_H | A |
ANGLE_L | NA | P
```

```
Byte 0   STATUS  (RST, MAGL, MAGH, MD, OVF)
Byte 1   ANGLE_H (bits 13..8)
Byte 2   ANGLE_L (bits 7..0)
```

STATUS and ANGLE are latched together. The read clears STATUS.RST (page 33).

40.2 Why read STATUS every cycle

- RST tells the host that CONF2 has been cleared, so the filter is off and DIR is 0 (page 58). The host must rewrite CONF2 in the same step.
- MAGL, MAGH and MD tell the host whether the angle is valid (page 52).

40.3 Host pseudocode

```
    every control period (1 ms):
        status, angle = burst_read(0x0B, 3)
        if status & 0x01:                 # STATUS.RST
            write(0x08, CONF2_VALUE)      # restore FILT, DIR, HYST, WD
        if (status & 0x06) or not (status & 0x08):
            hold_last_command()           # angle not valid (page 52)
        else:
            run_controller(angle)
```

40.4 Burst read of 4 bytes or more

A burst read that continues past ANGLE_L reads RAW_H and RAW_L (0x0E-0x0F). A burst read past 0x0F wraps incorrectly on all silicon revisions; see errata E-4 on page 59.

41 Address auto-increment and wrap

41.1 Auto-increment

The register pointer increments after every byte in both read and write transactions.

41.2 Wrap rules

Pointer after	Next pointer
0x0F	0x0B by design; 0x10 on all silicon (errata E-4)
0x1C	0x00
0x30	0x30 (CMD does not auto-increment)
reserved	next address

41.3 Intended wrap after 0x0F

The output block 0x0B-0x0F was designed to wrap back to STATUS (0x0B) so that a host could read STATUS, ANGLE and RAW continuously in one long burst. On all silicon the pointer moves to 0x10 instead (errata E-4, page 59), and the byte read from 0x10 is undefined.

41.4 Recommended practice

- Never rely on wrap. Start every transaction with an explicit pointer.
- Keep burst reads within 0x0B-0x0F.
- For long logging captures, issue one 5-byte burst read from 0x0B per sample.

41.5 Write auto-increment across CONF2

A multi-byte write starting at 0x06 runs through the reserved address 0x07 and into CONF2 at 0x08. Avoid writes that span 0x06-0x08; an unintended byte in CONF2 changes FILT, DIR, HYST or WD.

42 Bus error recovery

42.1 Stuck bus

If the host is reset in the middle of a read, the device may hold SDA low while it waits for clock pulses to finish a byte. To recover:

1. Configure SCL as a GPIO output and SDA as an input.
2. Toggle SCL up to 9 times until SDA reads high.
3. Generate a STOP condition.
4. Return the pins to the I²C peripheral.

42.2 NACK on the address byte

A NACK on the address byte means the device is not responding. Causes:

- The device is in reset (power-on, brown-out, watchdog, soft reset). It responds again 1.5 ms after the reset (page 11).
- The ADDR pin changed state before the last reset (page 13).
- TEST is not tied to GND (page 9).

After the device responds again, read STATUS. If RST = 1, rewrite CONF2.

42.3 Wrong data

If CHIP_ID does not read 0x14, the bus is corrupted or a different device answers. Check pull-ups and cable length (page 17).

42.4 A reset during a transaction

A reset that occurs in the middle of a transaction aborts it. The host sees a NACK or a byte of 0xFF. The next successful read of STATUS shows RST = 1.

42.5 Recovery is not configuration

Bus recovery restores communication. It does not restore CONF2. After any recovery, read STATUS and rewrite CONF2 if RST = 1 (page 45).

43 Power-up sequence

43.1 Sequence

t = 0	VDD rises through VPOR (2.8 V). Power-on reset.
t = 1.5 ms	I²C interface ready. STATUS = 0x01 (RST) plus field flags.
t = 2.0 ms	First valid angle. MD = 1 if a magnet is present.
t = 12 ms	AGC settled (10 ms time constant).

43.2 What the host should do

1. Wait at least 2 ms after VDD is stable.
2. Read CHIP_ID (expect 0x14) and REV.
3. Read STATUS. Expect RST = 1 on the first read after power-up.
4. Write CONF2 with the required FILT, DIR, HYST and WD values.
5. Read CONF2 back and compare.
6. Wait 12 ms for the AGC to settle before closing the control loop.
7. Start the control loop with the per-cycle burst read (page 40).

43.3 Slow supply ramps

If VDD rises slowly through VPOR and then dips (for example when a motor driver on the same rail is enabled), the device can see a power-on reset followed by a brown-out reset. The second reset clears the CONF2 value the host has just written. This is why step 7 must include the STATUS.RST check on every cycle, not only at start-up.

43.4 Host reset without encoder reset

If the host resets but the encoder does not, STATUS.RST is 0 on the first read and CONF2 still holds the old value. The host should write CONF2 anyway at start-up so that its configuration is known.

44 Default register values

Table 44-1. Register values after any reset.

Addr	Name	Value after reset	Source
0x00	CHIP_ID	0x14	fixed
0x01	REV	0xA1 or 0xA2	fixed
0x02	CONF1	OTP value (0x00 if blank)	OTP shadow
0x03	ZPOS_H	OTP value (0x00 if blank)	OTP shadow
0x04	ZPOS_L	OTP value (0x00 if blank)	OTP shadow
0x05	MPOS_H	OTP value (0x3F if blank)	OTP shadow
0x06	MPOS_L	OTP value (0xFF if blank)	OTP shadow
0x08	CONF2	0x00	volatile
0x0B	STATUS	0x01 plus field flags	volatile

44.1 CONF2 = 0x00 means

FILT = 0	the noise filter is off; output noise 12 LSB RMS (table 31-2)
DIR = 0	clockwise rotation increases the angle
HYST = 0	no hysteresis
WD = 0	watchdog off

44.2 "Reset" means every reset

The values in table 44-1 apply after a power-on reset, a brown-out reset, a watchdog reset and a soft reset. There is no reset source that preserves CONF2. Revision 1.0 of this manual implied that brown-out resets preserve CONF2. That was wrong; see errata E-3 on page 58.

44.3 Practical consequence

Every configuration value that the application needs and that is not in OTP must be written by the host after every reset. The host learns about resets from STATUS.RST (page 33).

45 Recommended initialisation

45.1 Start-up

```
    def encoder_init():
        wait_ms(2)
        assert read(0x00) == 0x14           # CHIP_ID
        read(0x0B)                          # clear RST from power-up
        write(0x08, CONF2_VALUE)            # e.g. 0x03 for FILT = 3
        assert read(0x08) == CONF2_VALUE
```

45.2 Every control cycle: rewrite CONF2 after a reset

```
    def control_step():
        status, angle = burst_read(0x0B, 3)  # page 40
        if status & STATUS_RST:              # bit 0, sticky, cleared by read
            write(0x08, CONF2_VALUE)         # the reset cleared CONF2
            log_event("encoder reset")
        ...
```

The rewrite costs one 3-byte write transaction, about 70 µs at 400 kHz. Because RST is cleared by the read that reports it, the rewrite must happen in the same control step. A host that only checks RST at start-up will run with FILT = 0 after the first brown-out.

45.3 Value to write

Choose FILT using page 49. For a robot joint with a 1 kHz loop and a crossover between 3 and 10 Hz, FILT = 3 is a good starting point (CONF2 = 0x03). Handle direction in host software if possible, so that a missed rewrite cannot reverse the feedback sign.

45.4 Firmware review checklist

- [ ] CONF2 is written at start-up, after CHIP_ID is confirmed.
- [ ] STATUS is read on every control cycle (burst read from 0x0B).
- [ ] CONF2 is rewritten in the same cycle whenever STATUS.RST = 1.
- [ ] Encoder resets are counted and logged.
- [ ] The angle is not used while MAGL or MAGH is set or MD is clear.
- [ ] No burst read runs past 0x0F (errata E-4).

46 OTP programming

46.1 What OTP stores

OTP holds CONF1, ZPOS and MPOS (page 29, page 30). After every reset these registers are reloaded from OTP. CONF2 is not stored in OTP (page 36).

46.2 Production procedure

1. Mount the magnet and complete mechanical assembly.
2. Supply VDD between 3.2 V and 3.6 V (errata E-1, page 57). On rev 0xA2 silicon 3.0 V is sufficient, but 3.3 V is recommended for all parts.
3. Read CHIP_ID (0x14) and REV. Record REV in the production log.
4. Move the joint to its mechanical zero and read RAW (0x0E-0x0F).
5. Write ZPOS = RAW. If the joint travels less than a full turn, move to the end stop, read RAW and write MPOS.
6. Write CONF1 if a non-default power mode or ADCSLOW is required.
7. Write 0x80 to CMD (0x30). Wait at least 10 ms without bus access.
8. Verify the burn as described on page 36.

46.3 When not to program OTP

Many robot designs keep ZPOS and MPOS at their defaults and apply the zero offset in host firmware. This avoids a production step that can be done only once and lets a joint be re-zeroed after repair. The configuration that matters for servo loops, CONF2, cannot be programmed in OTP in any case and must be written by the host after every reset.

46.4 Failure modes

- Burn at VDD below 3.2 V on rev 0xA1: partial burn, ZPOS and MPOS read back with random bits (errata E-1).
- Bus access during the 10 ms burn: the burn completes, but the access is NACKed and must be repeated.
- A second burn command is ignored. There is no way to erase OTP.

47 Servo loops: overview

47.1 Typical joint loop

A typical robot joint runs a cascaded loop on a microcontroller:

- Position loop: 1 kHz, input = encoder angle, output = velocity command.
- Velocity loop: 1 kHz, input = velocity estimate from the encoder angle, output = current command.
- Current loop: 10 kHz to 20 kHz in the motor driver.

The EncoderX-14 provides the angle for the outer two loops. The host reads the angle once per control period with a 3-byte burst read from 0x0B (page 40), which also returns STATUS.

47.2 What the encoder adds to the loop

- Delay: the read latency of 2 ms (page 12) for a host that polls once per 1 ms period and uses the result in the following period.
- Filter lag: the FILT setting adds the lag of a first-order filter with time constant tau = 2^n / 8000 s (page 31).
- Noise: 12 LSB RMS at FILT = 0, less with FILT > 0 (table 31-2).
- Quantisation: 1 LSB = 360 / 16384 = 0.022°.

47.3 Loop design summary

For a joint with a crossover between 3 Hz and 10 Hz:

- Delay and filter lag together must stay small compared with the phase margin. Table 12-1 gives the total lag at 4 Hz.
- Noise must be small enough that the derivative path does not drive the motor with noise (page 48).
- FILT = 3 balances these two needs (page 49).

The remaining pages of this part give the details.

48 Quantisation and noise in a derivative term

48.1 Why the derivative path matters

A PD or PID position loop, and every velocity loop, uses the difference of two successive angle samples. Differencing multiplies white noise by sqrt(2) and divides it by the sample period. At a 1 kHz loop rate, small angle noise becomes large velocity noise.

48.2 Velocity noise versus FILT (1 kHz loop)

Table 48-1. Noise of the one-step difference (angle[k] - angle[k-1]) / 1 ms.

FILT n	Angle noise (LSB RMS)	Velocity noise (LSB per ms RMS)	Velocity noise (° / s RMS)
0	12.0	17.0	373
1	6.93	9.8	215
2	4.54	6.4	141
3	3.10	3.6	78
5	1.51	0.8	17
7	0.75	0.2	4

For FILT >= 3 successive 1 ms samples are correlated by the filter, so the velocity noise falls faster than the angle noise.

48.3 Consequences at FILT = 0

- A derivative gain that is reasonable for the mechanics multiplies 373 °/s RMS of noise into the current command.
- The motor current carries broadband noise. The motor heats, the joint buzzes audibly, and a light link can resonate.
- The error appears as an increase in tracking error and in RMS motor current, not as a fault flag. Nothing in STATUS reports it.

48.4 Quantisation

At 16384 counts per turn and a 1 kHz loop, one LSB per sample corresponds to 22 °/s. Quantisation is small compared with the noise at FILT <= 3 and can be ignored for loop design.

49 Choosing the FILT setting

49.1 The trade-off

A larger FILT lowers the noise (table 31-2, table 48-1) and adds phase lag (table 12-1). The right value depends on the loop crossover frequency.

49.2 Phase lag at the crossover

Table 49-1. Total encoder phase lag (2 ms read delay plus FILT lag).

FILT n	tau	Lag at 4 Hz	Lag at 10 Hz
0	off	2.9°	7.2°
3	1 ms	4.3°	10.8°
5	4 ms	8.6°	21.3°
7	16 ms	24.8°	52.4°

The values at 4 Hz match table 12-1.

49.3 Recommendation for robot joints

For a joint with a 1 kHz loop and a crossover between 3 Hz and 10 Hz:

- FILT = 3 (CONF2 = 0x03) is the recommended starting point. It reduces the angle noise by a factor of 3.9 and the velocity noise by a factor of almost 5, and adds about 1.4° of lag at 4 Hz.
- FILT = 0 is not recommended. The loop is stable, but the derivative path drives the motor with noise (page 48).
- FILT >= 5 is only for slow loops. FILT = 7 costs about 25° at 4 Hz and can make a loop that was stable with FILT = 0 oscillate.

49.4 FILT must be maintained

The FILT setting chosen here is lost on every reset, including brown-out resets that the host does not otherwise see (page 11, page 58). Choosing FILT is half of the job; the other half is restoring it whenever STATUS.RST = 1 (page 45).

50 Velocity estimation

50.1 No velocity output

The device has no velocity register. The host estimates velocity from successive angle samples.

50.2 Unwrapping

The angle wraps from 16383 to 0. Before differencing, unwrap:

```
    d = angle[k] - angle[k-1]
    if d > 8192:   d -= 16384
    if d < -8192:  d += 16384
    velocity = d x (2 x pi / 16384) / T      # rad/s, T = loop period
```

This is valid while the joint turns less than half a revolution per loop period (30000 rpm at 1 kHz).

50.3 Estimators

Estimator	Noise	Lag	Note
One-step difference	highest	T / 2	Use with FILT >= 3
N-step difference	/ N	N x T / 2	Simple, good at low speed
Tracking observer	tunable	tunable	Best, needs tuning

50.4 Interaction with FILT

FILT filters the angle before the host sees it, so it also filters every velocity estimator that uses the angle. If CONF2 is cleared by a reset, FILT drops to 0 and the velocity noise rises by a factor of about 5 at 1 kHz (table 48-1) without any change in host code. The velocity estimate is often the first signal where a lost CONF2 becomes visible in logs.

50.5 Direction

If the host relies on DIR = 1 (page 32), a reset returns DIR to 0 and the sign of the velocity estimate reverses. Handle direction in host software where possible (page 15).

51 Diagnostics overview

51.1 What the device reports

Signal	Where	Meaning	Page
RST	STATUS bit 0	A reset occurred (sticky)	33, 53
MAGL	STATUS bit 1	Field too weak	52
MAGH	STATUS bit 2	Field too strong	52
MD	STATUS bit 3	Magnet detected	52
OVF	STATUS bit 4	CORDIC overflow (sticky)	52
AGC	0x1A	Front-end gain	35, 52
MAG	0x1B-0x1C	Field magnitude	35, 52
NACK	I²C	Bus or device fault	54

51.2 Reset sources and their evidence

All four reset sources (power-on, brown-out, watchdog, soft reset, page 11) have the same effect and the same evidence: STATUS.RST = 1 on the next STATUS read. The device does not record which source caused the reset. A host can narrow it down:

- Soft reset: the host itself wrote 0xA5 to CMD.
- Power-on: the host also restarted, or the encoder supply was switched.
- Watchdog: only possible with CONF2.WD = 1 (page 54).
- Brown-out: everything else. In a running joint this is the most common cause (page 53).

51.3 What the device does not report

- It does not report that CONF2 has changed. The only evidence is RST.
- It does not report excessive noise. Noise is a property of the FILT setting and the field, not a fault.
- It does not report a stalled host or a missed read.

52 Magnet field faults

52.1 Field flags

Condition	Flags	Angle valid
Field 30 mT to 90 mT	MD = 1	yes
Field below 30 mT	MAGL = 1, MD = 1	no
Field above 90 mT	MAGH = 1, MD = 1	no
Field below 20 mT	MAGL = 1, MD = 0	no
Field above 120 mT	MAGH = 1, MD = 0	no

MAGL, MAGH and MD are not sticky; they reflect the field at the time of the read. OVF is sticky and cleared by a STATUS read.

52.2 Typical causes

- Air gap too large (field too weak): see the air gap table on page 20.
- Air gap too small or wrong magnet grade (field too strong).
- Magnet loosened or cracked: MAG changes over time, AGC drifts.
- External field from a nearby motor: AGC and MAG change with motor current.

52.3 Host response

When the angle is not valid, hold the last motor command or bring the joint to a safe state. Do not use the angle (page 40). Log MAG and AGC with the event.

52.4 Trend monitoring

AGC moves slowly as the magnet ages and with temperature (page 35). A change of more than 20 % from the value recorded at production test indicates a mechanical problem.

52.5 Power-up glitch on rev 0xA1

On rev 0xA1 silicon MAGH can be set for up to 1 ms after a power-on reset even with a correct field (errata E-2, page 57). Wait 2 ms after reset before acting on MAGH.

53 Brown-out events in the field

53.1 How they happen

In a robot joint the encoder often shares the 3.3 V rail with the logic that drives the motor. A step in motor current can pull the rail below VBOR = 2.6 V for a few microseconds. A dip longer than the brown-out filter time tBOR (typ. 2 µs, page 11) resets the device.

53.2 What the host sees

- No bus error. The I²C interface is ready 1.5 ms after the reset; a read that falls inside that window is NACKed, later reads succeed.
- The angle keeps arriving. It is valid again 2 ms after the reset.
- STATUS.RST = 1 on the next STATUS read, then 0 again.
- CONF2 = 0x00: FILT = 0, DIR = 0, HYST = 0, WD = 0.

The joint keeps moving, but with the filter off. Velocity noise rises by a factor of about 5 (table 48-1). Tracking error and motor current rise, the joint may buzz. If the host relied on DIR = 1, the feedback sign reverses.

53.3 Detecting a brown-out

The only reliable evidence is STATUS.RST. Read STATUS on every control cycle with the 3-byte burst read from 0x0B (page 40) and test bit 0. A host that reads STATUS only at start-up will not detect a brown-out.

Indirect evidence, useful when analysing logs after the fact:

- A step increase in angle noise or velocity noise with no mechanical change.
- A single NACK on the encoder address during a motor current step.
- A coincidence between motor current steps and the change in noise.

53.4 Required response

In the same control step in which RST = 1 is read, rewrite CONF2 (page 45, errata E-3 on page 58). Count and log the event. If the count rises over time, improve supply decoupling (page 17); do not rely on decoupling alone.

53.5 Frequency

Field data from robot joints with shared supplies shows brown-out rates from less than one per year to several per hour, depending on layout and motor current. A design must assume that brown-outs happen.

54 Watchdog and bus faults

54.1 Watchdog

With CONF2.WD = 1 the device resets itself if the internal oscillator stalls for more than 1 ms. The watchdog reset has the same effect as every other reset: CONF2 = 0x00 and STATUS.RST = 1. Because WD lives in CONF2, the watchdog is disabled again after the reset until the host rewrites CONF2.

54.2 Bus faults

Symptom	Likely cause	Action
NACK on address	device in reset or not powered	retry next cycle
NACK on data	soft reset write (expected)	none
SDA held low	interrupted transaction	bus recovery
Wrong data after 0x0F	burst ran past 0x0F (E-4)	limit burst
Repeated NACK	wiring, ADDR pin, supply	stop, report

Bus recovery is described on page 42.

54.3 Host-side timeout

The host should treat more than three consecutive failed reads as an encoder fault and bring the joint to a safe state. A single NACK is normal after a reset and must not trip the joint.

54.4 After a bus fault

After any bus fault that lasted longer than 1.5 ms, read STATUS and check RST before using the angle. The fault may have been a reset.

55 Reliability data

55.1 Failure rate

Table 55-1. Reliability (JESD85 method, 60 % confidence).

Parameter	Condition	Value	Unit
FIT	55 °C junction	4.1	per 10^9 h
MTTF	55 °C junction	2.8e7	h
Early life failure rate		< 50	ppm

55.2 OTP data retention

Condition	Retention
125 °C	10 years
85 °C	> 50 years

55.3 Magnet ageing

The device itself does not age measurably. NdFeB magnets lose about 0.1 % of remanence per 1000 h at 100 °C. Over a product life this changes the field by a few percent, which the AGC compensates (page 35, page 52).

55.4 Reliability and resets

Brown-out resets do not damage the device and do not count against its reliability. They affect the application through the loss of CONF2 (page 58), not through wear.

56 Qualification and ESD

56.1 Qualification summary

Table 56-1. Qualification tests.

Test	Standard	Condition	Result
HTOL	JESD22-A108	125 °C, 3.6 V, 1000 h	0 / 231
Temperature cycling	JESD22-A104	-55 °C to 150 °C, 1000x	0 / 231
HAST	JESD22-A110	130 °C, 85 % RH, 96 h	0 / 231
High-temperature storage	JESD22-A103	150 °C, 1000 h	0 / 77
Latch-up	JESD78	±100 mA, 85 °C	pass
MSL	J-STD-020	260 °C reflow	MSL 3

56.2 ESD ratings

Model	Standard	Rating
Human body model (HBM)	JS-001	±2 kV
Charged device model (CDM)	JS-002	±500 V

56.3 Handling

Handle the device at an ESD-protected workstation (page 4). The SDA, SCL and ADDR pins are not protected against system-level ESD; add external protection if the I²C bus leaves the board.

56.4 Supply transients

The device tolerates supply transients within the absolute maximum ratings (page 9). Transients that pull VDD below VBOR cause a brown-out reset (page 11); they do not damage the device.

57 Errata summary

57.1 Errata list

Table 57-1. Known errata.

ID	Title	0xA1	0xA2	Page
E-1	OTP burn requires VDD >= 3.2 V	yes	no	57
E-2	MAGH glitch after power-on	yes	no	57
E-3	CONF2 cleared by every reset	yes	yes	58
E-4	Burst read wraps 0x0F -> 0x10	yes	yes	59
E-5	Low-power entry delay	yes	no	59

57.2 E-1: OTP burn requires VDD >= 3.2 V

Description: on rev 0xA1 an OTP burn at VDD below 3.2 V can program some bits incompletely. ZPOS and MPOS then read back with random errors.

Workaround: burn OTP only at VDD between 3.2 V and 3.6 V (page 36, page 46). Corrected in rev 0xA2.

57.3 E-2: MAGH glitch after power-on

Description: on rev 0xA1, STATUS.MAGH can read 1 for up to 1 ms after a power-on reset even when the field is correct.

Workaround: ignore MAGH for 2 ms after any reset (page 52). Corrected in rev 0xA2.

57.4 How to read the errata

E-3 is the erratum with the largest effect on servo applications. It is not a silicon defect that a later revision can fix; it is a property of the reset design that revision 1.0 of this manual described incorrectly. Every application must apply the E-3 workaround.

58 Errata E-3: CONF2 cleared by every reset

58.1 Description

Every reset clears CONF2 to 0x00. This includes power-on, brown-out, watchdog and soft reset (page 11). After the reset:

- FILT = 0: the noise filter is off (12 LSB RMS output noise).
- DIR = 0: the angle increases for clockwise rotation.
- HYST = 0 and WD = 0.

Revision 1.0 of this manual stated that brown-out resets preserve CONF2. That statement was wrong. There is no reset source that preserves CONF2.

58.2 Affected silicon

All revisions (0xA1 and 0xA2). This behaviour will not change.

58.3 Effect on applications

A brown-out reset is silent (page 11, page 53). A host that writes CONF2 only at start-up runs with FILT = 0 after the first brown-out. In a servo loop:

- Angle noise rises from 3.10 LSB RMS (FILT = 3) to 12.0 LSB RMS.
- Velocity noise at 1 kHz rises from 78 °/s RMS to 373 °/s RMS (table 48-1).
- Motor current noise, tracking error and audible noise increase.
- If DIR = 1 was used, the feedback sign reverses.

The configuration read back after the original write was correct, so a start-up self-test does not catch this failure.

58.4 Workaround

Read STATUS on every control cycle and rewrite CONF2 in the same cycle whenever STATUS.RST = 1:

```
    status, angle = burst_read(0x0B, 3)
    if status & 0x01:                # STATUS.RST, sticky, cleared by read
        write(0x08, CONF2_VALUE)     # restore FILT (and DIR, HYST, WD)
```

The rewrite must happen in the same cycle because the read that reports RST also clears it (page 33). See page 45 for the full initialisation.

58.5 Verification

Force a soft reset (write 0xA5 to CMD) while the loop is running. Within one control cycle CONF2 must read back as CONF2_VALUE.

59 Errata E-4 and E-5

59.1 E-4: burst read wraps 0x0F -> 0x10

Description: the register pointer was designed to wrap from 0x0F (RAW_L) back to 0x0B (STATUS). On all silicon it continues to 0x10 instead. The byte read from 0x10 is undefined (page 41).

Affected silicon: 0xA1 and 0xA2.

Effect: a host that reads more than 5 bytes from 0x0B, expecting STATUS again at byte 6, reads an undefined value instead. If that value has bit 0 set, the host may treat it as a reset; if it has bit 0 clear, a real reset reported in a later read can be missed by code that only checks byte 6.

Workaround: keep burst reads within 0x0B-0x0F. Start every read transaction with an explicit register address. The recommended read is the 3-byte burst from 0x0B (page 40).

59.2 E-5: low-power entry delay

Description: on rev 0xA1, after CONF1.PM is changed from 0 to 1 or 2, the device takes up to 20 ms instead of the specified 1 ms to enter the low-power mode. During that time the current consumption is that of active mode.

Affected silicon: 0xA1. Corrected in 0xA2.

Effect: battery-powered designs that switch to low power for short idle periods save less energy than expected. Servo loops, which should use PM = 0 (page 29), are not affected.

Workaround: allow 20 ms for low-power entry on 0xA1, or use rev 0xA2.

60 Glossary and licence

60.1 Glossary

AGC	Automatic gain control of the magnetic front end (page 35)
BOR	Brown-out reset: reset caused by VDD falling below VBOR
CONF2	Volatile configuration register 0x08 (FILT, DIR, HYST, WD)
CORDIC	Algorithm that computes the angle from the sine and cosine
FILT	CONF2 bits 2..0: output noise filter setting (page 31)
LSB	One count of the 14-bit angle, 360 / 16384 = 0.022°
OTP	One-time programmable memory (page 36)
POR	Power-on reset
RST	STATUS bit 0: a reset occurred since the last STATUS read
tau	Time constant of the FILT filter, 2^n / 8000 s
VBOR	Brown-out reset threshold, 2.6 V typical
VPOR	Power-on reset threshold, 2.8 V typical

60.2 Fictional device notice

The EncoderX-14 is a fictional device. This manual was written for the ChronoSight project as a test corpus. Any similarity to a real product is unintended. Do not use the values in this manual to design real hardware.

60.3 Licence

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